

River Deep Mountain AI

Model Benchmarking Report: *E. coli* (BB1B)

Independent Model Benchmarking Report - *E. coli* (BB1B)

FINAL REPORT

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Glossary

AI	Artificial Intelligence
CFU	Colony Forming Unit
CORINE	Coordination of Information on the Environment
DO	Dissolved Oxygen
EA	Environment Agency
FN	False Negative
FP	False Positive
MAE	Mean Absolute Error
ML	Machine Learning
MLGA	Membrane Lactose Glucuronide Agar
MODIS	Moderate Resolution Imaging Spectroradiometer
RMSE	Root Mean Square Error
WIMS	Environment Agency Water Information Management System

Summary

The benchmarking of the RDMAI Open *E. coli* model focused on a model with a similar approach (see **Model Approach** in table below – both models use an AI/ML approach) developed by Wessex Water for Warleigh Weir. However, since the Wessex Water model is for only a single site, the benchmarking additionally examined a citizen science dataset of *E. coli* measurements for Great Britain. While the citizen science dataset is not a model, but rather a dataset of monitored *E. coli*, including this dataset in the benchmarking is justified. This is because the aim of the benchmarking extends beyond assessing model accuracy, to determining whether the RDMAI Open *E. coli* model can reduce costs and improve safety ('safety' alludes to the possibility of gastro-intestinal illness in humans due to coming into contact with bathing waters contaminated by faecal matter, and consequently faecal pathogens).

Model performance

Benchmarking against the Warleigh Weir Wessex Water Model

It should be mentioned that benchmarking the national-level RDMAI *E. coli* model against the site-specific Warleigh Weir Wessex Water model is a considerable limitation - it is to be expected that the Wessex Water model will show far better performance for Warleigh Weir.

For Warleigh Weir, the RDMAI Open *E. coli* estimator models have an R^2 of -0.286 to -0.085 (RMSE and MAE of between 2752–3110 and 1465–1812, respectively). This is compared to the R^2 of the Warleigh Weir model of 0.2 (RMSE and MAE of 3907 and 1702, respectively).

False negatives, when applying a benchmark threshold of 500 CFU/100 ml to the light and advanced RDMAI Open *E. coli* estimator results for Warleigh Weir, were 51% and 66%, respectively; those for the classifier were 61% and 37%, respectively. This is compared to the false negatives of the Warleigh Weir model of 0% when using either a 500 CFU/100 ml or 1,000 CFU/100 ml threshold.

Benchmarking against the Citizen Science dataset

The R^2 of the RDMAI Open *E. coli* light estimator was -0.73 (RMSE and MAE of 1250 and 761, respectively). False negatives, when applying a benchmark threshold of 500 CFU/100 ml to the light estimator results for the citizen science dataset was 26%; that for the light classifier was 19%. False positives range between $< 1\%$ and 57%

The average model performance (in terms of false negatives) for all sites nationally (in terms of the citizen science dataset) improves compared to the performance when applied to a single site (Warleigh Weir).

Other aspects of the model are summarised in the table below

Model Aspects	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
Model Approach	The Wessex Water Warleigh Weir models are AI/ML models, trained using water quality and rainfall data. There are two variants: • Regressor: gives <i>E. coli</i> concentrations in CFU/100 ml • Classifier: gives a binary lower risk/higher risk prediction, trained on thresholds of 900-1000 CFUs/100 ml The results of the classifier are used in an online traffic-light system-based tool to inform bathing water safety at Warleigh Weir.	The RDMAI Open <i>E. coli</i> models are AI/ML models. There are four variants, based on: • Training data: ○ The light model: requires land use, weather, and remote sensing data ○ The advanced model: requires all the data used by the light model plus certain key water quality parameters. • Model output: ○ The estimator: provides <i>E. coli</i> concentrations in CFU/100 ml ○ The classifier: provides a binary output of lower risk or higher risk for bathing. For example, 0 and 1, representing lower and higher risk, respectively.
Applications	The purpose of the Warleigh Weir model is to inform bathing water safety. Wessex Water uses the results of the classifier to advise the public, in near real time, of the risks of bathing at Warleigh Weir.	Given the expense of traditional laboratory <i>E. coli</i> measurements and the associated temporal and spatial constraints, these models can support more comprehensive assessment of bathing water safety. Although both the validation and benchmarking exercises revealed that the models at this stage are not

Model Aspects	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
		sufficiently accurate to be used on their own, they can act as additional sources of information to supplement pollution risk warnings and traditional monitoring, such as regulator, water company and citizen science <i>E. coli</i> monitoring. There is potential for near real-time application of the light RDMAI Open <i>E. coli</i> model, given that the required input data – MODIS remote sensing data and weather data, can be measured in near real time. Near real-time application of the advanced RDMAI Open <i>E. coli</i> model is less likely, given delays in measuring the additional supporting water quality parameters, such as phosphate, which is mostly still measured using slow laboratory methods.
Geographic Constraints	The Wessex Water <i>E. coli</i> models are specific for Warleigh Weir.	The RDMAI Open <i>E. coli</i> advanced model has been trained on England-specific data, whereas the light model has been trained on data for Great Britain. Provided the input data are available, they can be applied outside of these areas, but the results given would be questionable without retraining. The light model can be applied at any point along a river or coastal water body, whereas the advanced model can only be applied at points with supporting water quality data. Light model: • Great Britain, at any point. Advanced model: • England, at points at which supporting water quality data are available.

Model Aspects	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
Temporal Resolution	The Wessex Water Warleigh Weir model runs at a half-hourly timestep.	The temporal resolution of the light model is determined by weather and remote sensing data, at a 1-day resolution or finer. Light model: • 1-d resolution or finer The temporal resolution of the RDMAI Open <i>E. coli</i> advanced model is determined by the temporal resolution of the supporting water quality input data. Given that some parameters, such as phosphate-P, cannot be measured by sondes, temporal resolution will be limited. Advanced model: • Determined by supporting water quality data: typically, one or two virtual sample points per month.
Data Inputs Required	The Wessex Water Warleigh Weir model requires rainfall, flow and water quality data. Water quality data parameters needed are: • Ammonium • Ammonia • Chloride	For the RDMAI Open <i>E. coli</i> light models, data requirements are: • CORINE land cover • Open Meteo API weather data • MODIS remote sensing data

Model Aspects	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
	• Conductivity • DO • pH • Suspended solids • Water temperature • Turbidity	All these data are freely available, except for the Open Meteo weather data, which requires a subscription if the user is a commercial entity. The RDMAI Open <i>E. coli</i> advanced models require water quality measures in addition to the data mentioned above. These include: • Water temperature • Suspended solids • Ammoniacal-N • Nitrate-N • Nitrite-N • Phosphate-P
Model Performance	The Wessex Water Warleigh Weir regressor (estimator) has an R^2 of 0.2. Using the estimator results, the model gives no false negatives when applying a benchmark threshold of 500 CFU/100 ml, but it does have a high false positive rate of 95%.	It should be mentioned that benchmarking the national-level RDMAI <i>E. coli</i> model against the site-specific Warleigh Weir Wessex Water model is a considerable limitation - it is to be expected that the Wessex Water model will show far better performance for Warleigh Weir. The RDMAI Open <i>E. coli</i> estimator models have an R^2 of -0.085 to -0.73.

Model Aspects	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
	The classifier, using Wessex Water's own thresholds (900 to 1000 CFU/100 ml), provides a false positive rate of 48% and no false negatives.	False negatives, when applying a benchmark threshold of 500 CFU/100 ml to the light and advanced estimator results for Warleigh Weir, were 51% and 66%, respectively; those for the classifier were 61% and 37%, respectively. False negatives, when applying a benchmark threshold of 500 CFU/100 ml to the light estimator results for the citizen science dataset was 26%; that for the light classifier was 19%. False positives for the light and advanced estimator results for Warleigh Weir were 25% and 10%, respectively; those for the classifier were <1% and 5%, respectively. False positives for the light estimator and light classifier results for the citizen science dataset were 57% and 14%, respectively. The average model performance (in terms of false negatives) for all sites nationally (in terms of the citizen science dataset) improves compared to the performance when applied to a single site (Warleigh Weir).
Cost & Time	The model is not open – it is an operationalised model within Wessex Water. Water quality data used in the model are sampling and sonde data collected by Wessex Water – therefore, it is difficult to provide estimates for cost of running the model. Communication with Wessex Water indicated that most of the time and effort taken to operationalise the model relates to bringing in real-time data, rather than due to technicalities around machine learning.	For commercial organisations, a professional subscription is needed to access Open Meteo API historical weather data. This currently costs €99/month. For non-commercial organisations, the API can be used freely. All other data used in the model are freely available. Extracting the land use, weather and remote sensing data is automated, but can take time, depending on the dataset length. Licence costs:

Model Aspects	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
		• Open Meteo API professional subscription for commercial users: €99/month. Time: • Extracting, processing and running the model for around 2000 data points: 4 hours, provided the model user is experienced and proficient.
Strengths	The model has a very low false negative rate (0%) – the model indicates periods of higher risk bathing at Warleigh Weir that directly match the observations of <i>E. coli</i>. This makes the model suitable for use for predicting bathing water safety.	• The model is freely available, both to use and adapt/extend/retrain. • The input data are mostly freely available. • The advanced and light models can be applied to England and Great Britain, respectively, with potential to expand the model further. • The data inputs required allow potential for the light model to be used to support real-time decision making or future scenario analysis. • The model can be re-trained to simulate intestinal enterococci, in line with the EA PRF system.
Limitations	The model is also conservative, in that it often predicts unacceptable bathing water quality when the observations indicate otherwise. The model is restricted to application to Warleigh Weir.	• The models at this stage show relatively limited accuracy. Validation highlighted a tendency of the model to be biased towards the majority class (low <i>E. coli</i> measures), which results in under-simulation of high <i>E. coli</i> and increased rates of false negatives. This can be improved through continual development using additional data or improvements to the model.

Model Aspects	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
	The model is not open – it is a model that has been operationalised by Wessex Water to inform bathing water safety at Warleigh Weir.	• Within the extraction of land use and other data, the models at this stage do not delineate upstream contributing catchment but rather work according to a circular buffer around a point. This shortcoming can be improved in further development. • Training <i>E. coli</i> concentration data for the RDMAI Open <i>E. coli</i> model was capped at 10,000 CFU/100ml. This could have an impact on the model performance in predicting extreme <i>E. coli</i> concentrations. • The advanced model is dependent on certain key measured water quality parameters, such as phosphate-P. This dependence constrains both the spatial and temporal extents at which the model can be run. Dedicated sampling of these parameters, if not available in the EA WIMS, will incur additional costs.

1. Benchmarking Methodology

1.1 Benchmarking Model Selection

1.1.1 RDMAI Open *E. coli* Model Overview

Escherichia coli (*E. coli*) is used as a faecal indicator organism in water quality assessments. Therefore, the presence of *E. coli* in inland and coastal waters can indicate risk of human exposure to enteric pathogens from activities such as bathing. Because *E. coli* is typically measured using laboratory-based techniques, the spatial and temporal frequency of monitoring is limited. An alternative or complimentary tool to laboratory measurements of *E. coli* would be model simulation, which can overcome these limitations and inform real-time risk assessment. The aim of the RDMAI model is to predict *E. coli* concentrations to support bathing water quality assessment.

The RDMAI Open *E. coli* model comes in two variations, according to data requirements and output, respectively, resulting in four variants.

By data requirements:

- **The light model** requires land use data, weather data, and remote sensing MODIS data. As such, it can be applied anywhere in Britain (as the model was trained on *E. coli* datasets for Britain) and at a wide range of temporal resolutions.
- **The advanced model** requires all the data used in the light model, and additional water quality parameters, such as suspended solids, phosphate-P, and nitrate-N and nitrite-N. Importantly, the frequency and spatial positions of simulations by the advanced model are limited to the frequency and spatial positions of these additional water quality parameters. The model was trained on Environment Agency (EA) *E. coli* and water quality measures for England. As such, the model application is restricted to England and the spatial and temporal resolutions that the model can be applied are restricted to a large degree by those of the observed water quality data.

By model output:

- **The estimator** provides simulations of *E. coli* concentrations in Colony Forming Units (CFU)/100 ml.
- **The classifier** provides binary outputs: 'lower risk' or 'higher risk', based on thresholds of *E. coli* concentration associated with risk of illness in bathing waters. Threshold 1 aligns with national and international guidelines from the UK, US Environmental Protection Agency (EPA), and the World Health Organisation (WHO). Threshold 2 is based on the threshold used by the EA in coastal bathing waters.

-
- *Threshold 1: E. coli* > 1000 CFU/100mL = 'Higher Risk'; *E. coli* < 1000 CFU/100mL = 'Lower Risk'
 - *Threshold 2: E. coli* > 500CFU/100mL = 'Higher Risk'; *E. coli* < 500 CFU/100mL = 'Lower Risk'

E. coli concentration data used to train the RDMAI Open *E. coli* model was capped at 10,000 CFU/100ml. This limitation could affect the model's ability to correctly predict higher *E. coli* concentrations.

Model validation demonstrated shortcomings in RDMAI model performance against actual measured data. First, there is a bias in the model towards the majority class, resulting in a relatively high rate of false negatives (FNs) (15% and 27% for the best performing advanced and light models, respectively during validation). The classifier model also shows a sensitivity to the chosen threshold, which affects the FN/FP balance. Therefore, the model should not be used as an alternative to monitoring for bathing water quality assessment. Rather, the model should be considered a complimentary tool to enhance existing water quality assessment practices. For a more detailed example of how the model could be used operationally, please refer to the use-case documentation published in the project [Github](#).

More information on the Open *E. coli* model can be obtained from the [RDMAI Open E.coli Github site](#).

1.1.2 Overview of Existing Industry Model(s)/Tool(s)

EA's pollution risk forecasts / SWIMFO

The EA has developed a pollution risk forecasting (PRF) system to warn bathers of expected reductions in water quality resulting from short-term pollution events. Information on the PRF model was obtained from an informal call with the EA, rather than from published model documentation. In summary:

- The model predicts intestinal enterococci (working on a log scale) at bathing sites in England. It uses a threshold concentration of 63 CFUs/100 ml based on a human health risk – indicates a 70% probability of a 10% chance of human gastrointestinal illness.
- The models are based on bivariate regression during data exploration and are finally stepwise multiple regression models.
- Each bathing site has its own model, trained on local data.
- These are forecast models. Therefore, they can only use variables which can be forecast, including:

-
- Rainfall – based on radar rainfall
 - Wind
 - UV
 - Tide
 - From the forecast variables, additional variables can be derived (up to 40).
 - Antecedent totals are applied for some of the variables, like rainfall.
 - Depending on the site, variable values from the entire contributing catchment are used, or a simple 1-km radius is used to delineate variable values.
 - The models generate 10 hourly predictions of intestinal enterococci per day, with an average used to issue the daily warning – published on the website, or for subscribers, warnings sent by text or email – warning signs at some beaches are also put up.
 - Model training starts with data from as far back as 2004. However, the start dates for the models differ – want to account for improvements in water quality due to improved treatment, CSO capacity, etc.
 - While there are around 450 models in total (one for each bathing site), only models deemed accurate enough are used to forecast. This is determined through an overall compound goodness-of-fit score incorporating various metrics, including R^2 , percentage of the worst samples that are correctly simulated, etc. The models are also checked against information given by local experts.
 - The R^2 s of the models deemed good enough for forecasting typically range between 0.3 to 0.4.

The tool predictions are shown on an online tool [Swimfo](#), which also gives up to date classifications of bathing sites based on measured water quality. This information can help bathers choose where and when to swim.

We chose not to benchmark the RDMAI Open *E. coli* model against the EA PRF model for the following reasons:

- The models predict different variables: the EA PRF model predicts intestinal enterococci while the RDMAI Open *E. coli* model predicts *E. coli*.
- The model predictions of whether bathing sites are safe are based on different thresholds: the EA PRF model is based on an intestinal enterococci concentration

threshold of 63 CFU/100 ml while the RDMAI Open *E. coli* classifier models are based on *E. coli* concentrations of 500 or 1000 CFUs/100 ml.

- After downloading all available historic forecasts from the PRF for England (2013-2025), which included 50,000 lines of data, it is very difficult to tell which forecasts indicate a high bacterial load.

Wessex Water's *E. coli* simulation model

Wessex Water has developed a machine learning based *E. coli* model for Warleigh Weir. Warleigh Weir is a popular summer bathing spot along the River Avon near Bath. Wessex Water have operationalised the model for informing bathing water safety. They have developed two variants of the model:

- A regressor (estimator) which gives predictions of *E. coli* concentration (CFU/100 ml).
- A classifier, based on a threshold of 1000 CFU/100 ml, which Wessex Water has subsequently reduced to 900 CFU/100 ml.

The results of the classifier are public facing – published on a website informing bathing water safety at Warleigh Weir.

The model takes rainfall data and water quality data as input – both frequently measured sonde data and less frequently measured nutrient data, such as nitrate, nitrite and phosphate. The model is not an open model, with use restricted to the operations of Wessex Water.

This model was chosen for benchmarking given the limited availability of models predicting microbial water quality in England/Great Britain and because of its similarity in methodology (ML approach) and objective (informing bathing water safety) to the RDMAI Open *E. coli* model.

Citizen science measures of *E. coli*

While the Wessex Water *E. coli* model provides observations and simulations for *E. coli* of around 40,000 data points between 2021 and 2025, these are for a single location: Warleigh Weir. This limitation on spatial coverage limits the benchmarking potential with the Wessex Water data provided. The aim of the benchmarking exercise is not just to assess the accuracy of the RDMAI models against industry standard models, but also to assess whether the RDMAI models can reduce monitoring cost and improve risk assessment. Therefore, the decision was made to also benchmark the RDMAI Open *E. coli* model against citizen science measures of *E. coli*. These measures were taken from the [CITIZEN SCIENCE DATA HQ](#) site.

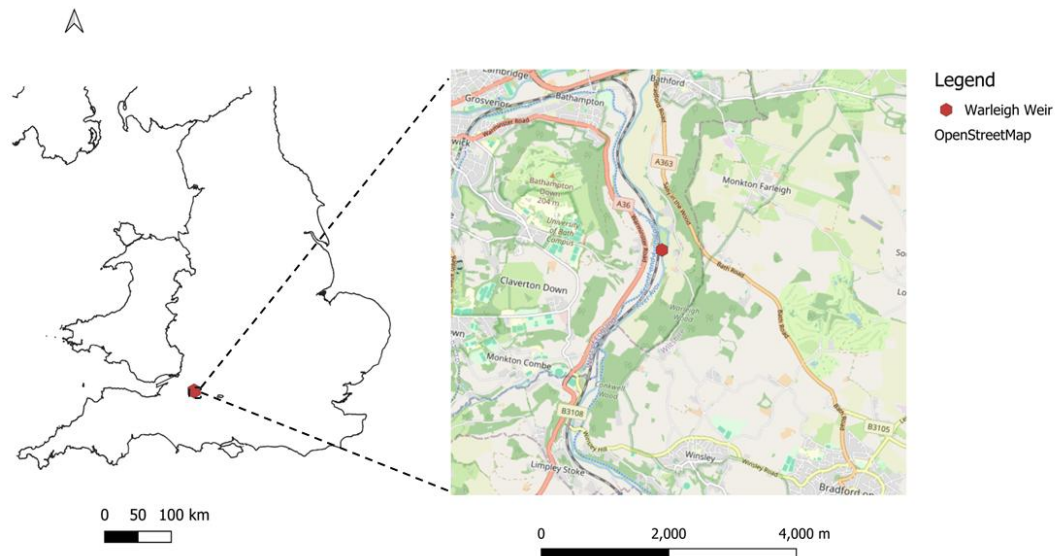
1.2 Catchment Selection

Wessex Water Warleigh Weir site

Figure 1.1 Shows the location of Warleigh Weir in South West England. A major constraint of benchmarking against the Wessex Water Warleigh Weir site is that it is a single site – the

performance of the RDMAI Open *E. coli* model at one site cannot be considered representative of its overall performance nationally.

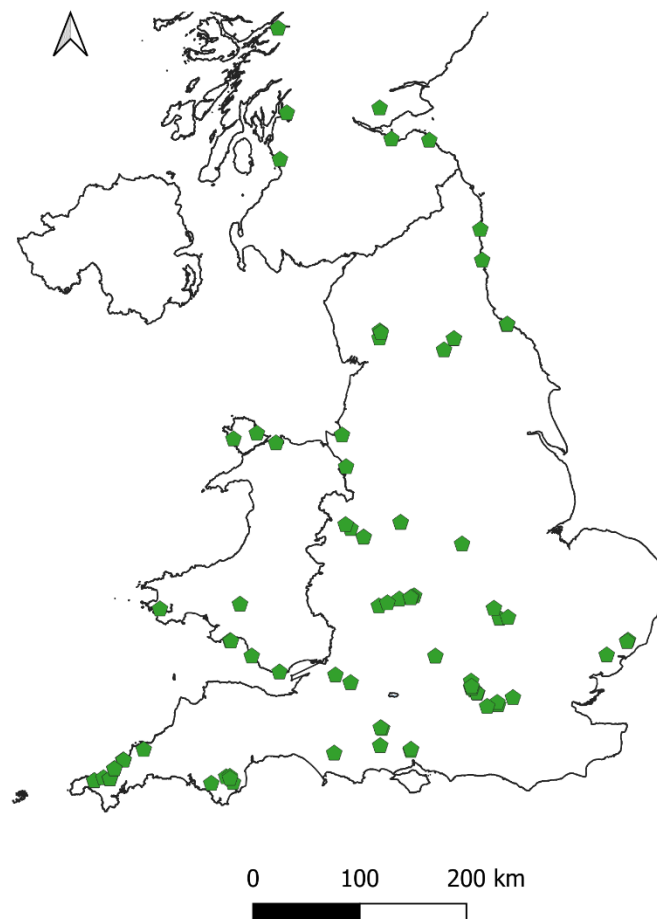
Figure 1.1 Map showing location of Warleigh Weir in South-West England as well as nearby Environment Agency water quality monitoring points



Citizen science measures of *E. coli*

Figure 1.2 shows the locations of citizen science *E. coli* monitoring points in England, Wales and Scotland. These data are appropriate for application to the RDMAI Open *E. coli* light models as these models have been trained with *E. coli* data from the English, Welsh and Scottish regulators. Because the citizen science *E. coli* data have minimal or no supporting water quality data, this dataset cannot be applied to the RDMAI Open *E. coli* advanced models.

Figure 1.2 **Locations of citizen science *E. coli* monitoring points in England, Wales and Scotland**



1.3 [Benchmarking Methodology](#)

[Wessex Water Warleigh Weir model](#)

Wessex Water provided a dataset of measured and simulated *E. coli* for a single site: Warleigh Weir. These measures and simulations are at a 30 min temporal frequency between 2021 and 2025, totalling around 40,000 individual data lines. However, within that dataset, 301 data lines contain laboratory measured *E. coli*, whereas the remainder is inferred from the sonde data. Therefore, the benchmarking was on a dataset containing 301 lines. Included in the data are measured and simulated *E. coli*, as well as water quality measurements, primarily ammonium (N), chloride, conductivity, dissolved oxygen (DO), pH, phosphate-P, suspended solids, temperature, and turbidity. Many of the data lines do not contain readings for suspended solids and phosphate-P. For example, approximately 300 of the 40,000 lines contain actual measures for suspended solids. The Wessex Water dataset also contains classification of *E. coli* risk to human health as binary variables. However, these classifications are based on different *E. coli* concentration thresholds than those applied in the RDMAI Open *E. coli* classifier models during model training. Therefore, this benchmarking exercise applies the threshold of 500 CFU/100 ml to the Wessex Water Warleigh Weir model regressor simulations, to be able to compare

performance metrics amongst the assessed models. But, when assessing the Wessex Water Warleigh Weir model classifier, the Wessex Water thresholds of 900 to 1000 CFU/100 ml apply. This is because Wessex Water provided the results of their classifier – we do not have access to their model and we cannot re-run their model with a different classification threshold.

Benchmarking of the RDMAI Open *E. coli* light models

The light versions of the RDMAI Open *E. coli* model (both estimator and classifier) do not require supporting water quality measures. Therefore, the light version could be applied to the entire Wessex Water dataset (301 data lines). Existing Python scripts (Jupyter Notebooks) within the RDMAI model GitHub Repository were adapted to prepare the data for running the light models. Primarily, this involved determining land use around Warleigh Weir (based on [CORINE](#) land cover), weather data for each data line (based on the [Open Meteo API](#)), and remote sensing estimates [[Moderate Resolution Imaging Spectroradiometer \(MODIS\)](#)]. Existing Python scripts (Jupyter Notebooks) within the RDMAI model Github repository for model validation were then adapted to run the prepared benchmarking dataset.

Benchmarking of the RDMAI Open *E. coli* advanced models

The RDMAI Open *E. coli* advanced models require all the data needed for the light models (described above), and the following water quality measurements: water temperature, suspended solids, ammoniacal-N (the sum of ammonia-N plus ammonium-N), nitrate-N, nitrite-N, and phosphate-P. The Wessex Water dataset contains ammonium-N for most data lines, and it was assumed that ammonium-N dominates ammoniacal-N [the balance between ionised and unionised ammonia is determined by temperature and pH; usually unionised ammonia-N constitutes less than 10% of ammoniacal-N under typical pH and water temperature of England's rivers ([EA, 2012](#))]. Most of the data lines in the Wessex Water dataset contained measures of water temperature. However, very few of the data lines contained actual readings of phosphate-P and suspended solids, and the Wessex Water dataset did not include nitrate-N and nitrite-N. Therefore, to run the RDMAI Open *E. coli* advanced models, missing data required to run the RDMAI Open *E. coli* advanced was infilled from a Wessex Water's water quality dataset for Warleigh Weir downloaded from the [Stream Portal](#).

This resulted in a total of 176 valid data lines with which to run the RDMAI Open *E. coli* advanced models.

Calculating performance metrics

RMSE (Root Mean Square Error): quantifies the average magnitude of the prediction error.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Where:

y_i = observed values

$\hat{y}_i$ = predicted values
 n = number of observations

R^2 (Coefficient of Determination): measures the proportion of the variance in the data that is explained by the model.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Where:

$\bar{y}$ = mean of the observed data

Mean Absolute Error (MAE): measures the average absolute differences between predicted and actual values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

The performance metrics refer to:

- False negatives (FNs): these are claims by the model of lower risk *E. coli* levels, when the water quality is at a higher risk level.
- False positives (FPs): these are claims by the model of higher risk *E. coli* levels, when the water quality is at a lower risk level.

For a model indicating human health risk, the FNs are a more important statistic, as they have a larger impact - false positives lead to possible inconvenience of bathers avoiding swimming when it was safe to swim; false negatives can lead bathers deciding to swim in contaminated water, putting their health at risk.

Citizen Science measures of *E. coli*

The citizen science measures of *E. coli* are available online at the [CITIZEN SCIENCE DATA HQ](#) site. Importantly, these are actual observations of *E. coli*, not model estimations. The data are visualised as points on an interactive map of the United Kingdom. Clicking on a point brings up a table of *E. coli* measures, one line per measure, along with some key supporting information, including waterbody, site, site type and date. These data were extracted and processed for input into the RDMAI Open *E. coli* models. The data provided do not include co-ordinates. Therefore, the co-ordinates were extracted from Google Earth based on the provided waterway and site names. The data were plotted in GIS and cross-checked with the online map to ensure that coordinates were correctly identified. The measurement dates in the data provided also do not provide the time of measurement. Therefore, a 12:00 time measurement was assumed for all points. The *E. coli* measures have inadequate supporting water quality data to run the RDMAI Open advanced model, therefore benchmarking was used with the

RDMAI Open light model only. Once the data were consolidated in a spreadsheet, all data lines with missing information, such as any of the weather parameters or MODIS indices, were deleted, as complete data lines are needed for the model to run correctly.

Benchmarking of the RDMAI Open *E. coli* light models

The RDMAI Open *E. coli* light model (estimator and classifier) was applied to the citizen science *E. coli* data as the light versions do not require supporting water quality measures. The existing scripts within the model GitHub Repository were adapted to prepare the data, as described in the process used for benchmarking against the Wessex Water Warleigh Weir data.

1.4 Benchmarking Limitations

RDMAI Open *E. coli* model

Training *E. coli* concentration data for the RDMAI Open *E. coli* model above 10,000 CFU/100 ml was capped to this value to ensure model focus on lower concentration levels. This could have an impact on the model performance if the benchmarking data contains many *E. coli* concentrations exceeding this threshold. Approximately 6% of the measures *E. coli* concentrations in the Warleigh Weir dataset exceeded 10,000 CFU/100, whereas in the citizen science dataset, this proportion was around 1%.

Wessex Water Warleigh Weir model

An obvious limitation of benchmarking the RDMAI Open *E. coli* models against the Warleigh Weir model is that the former is a national model, whereas the latter is a site-specific model. The Wessex Water model is trained, tested and validated against the Warleigh Weir site, but the RDMAI Open *E. coli* models have not previously been exposed to this site. The validation against a single site also does not give a robust estimation of the accuracy of the RDMAI Open *E. coli* models. An additional limitation with applying the RDMAI Open *E. coli* advanced models to the Warleigh Weir site was a reduction in data lines from 301 to 176 due to a lack of additional water quality parameters needed by the advanced models.

Citizen science measures of *E. coli*

The citizen science measures of *E. coli* do not have any supporting water quality data. Therefore, it is not possible to apply the RDMAI Open *E. coli* advanced models to these data. In addition, much of the *E. coli* data in this dataset is reported as greater than '>' and less than '<' values. In these cases, the threshold values reported have been taken as the measured *E. coli*. This has resulted in many observations being stated as 'greater than' 1000 CFU/100 ml, which were taken to be 1000 CFU/100 ml. While dates of measures are provided, the time of day is not indicated. This will have implications for weather data extracted for these monitoring points as weather shows intra-day variations. It was assumed that each sample was taken at 12:00.

Due to the nature of citizen science data, there may be uncertainty in regard to the accuracy of *E. coli* measurements – particularly concerning sample collection and record keeping. While

the samples are measured in a laboratory, citizens are given responsibility for preventing contamination of the samples during collection (see the [instructional video](#)) – there is no guarantee that cross-contamination of samples is completely avoided. This is a potential limitation to consider when evaluating the reliability of the dataset used to validate the RDMAI model.

2. Benchmarking Results

2.1 Model ‘Ease-of-Use’

Table 2.1 Comparison of model ‘ease-of-use’

Model Outputs	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
Installation & Run Process	<i>The Wessex Water E. coli models for Warleigh Weir are not open models – they are restricted for use within the operations of Wessex Water. As such, it is not possible to comment on the process to install and run the Wessex Water model.</i>	Installing and running the RDMAI Open <i>E. coli</i> model requires experience of setting up a Python Conda environment and basic skill in Python language and Jupyter Notebooks. This assumes that the trained model files are provided and the model does not need to be re-trained. Investigating the process for retraining the model was outside the scope of this report but would likely require advanced data science expertise. The RDMAI Open <i>E. coli</i> model repository on GitHub provides Jupyter Notebook scripts that can be adapted for preparing the dataset and running the light or advanced models (both estimator and classifier versions). Some of the provided scripts for data preparation needed to be slightly adapted to work on a Windows 11 machine. Time needed for extraction of weather information from the Open Meteo API and MODIS data is dependent on the dataset size; for the Wessex Water dataset with 40,000 measures, data extraction took 4 – 5 hours on a standard laptop and home internet connection. Once the

Model Outputs	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
		data files were prepared, running the trained model typically requires < 1 minute using the Jupyter Notebooks.
Data inputs	<i>The Wessex Water Warleigh Weir model requires rainfall, flow and water quality data. Water quality data parameters needed are:</i> • <i>Ammonium</i> • <i>Ammonia</i> • <i>Chloride</i> • <i>Conductivity</i> • <i>DO</i> • <i>pH</i> • <i>Suspended solids</i> • <i>Temperature</i> • <i>Turbidity</i>	For the RDMAI Open <i>E. coli</i> light models, data requirements are: • CORINE land cover • Open Meteo API weather data • MODIS remote sensing data All these data are freely available, except for the Open Meteo weather data, which requires a subscription if the user is a commercial entity. The RDMAI Open <i>E. coli</i> advanced models require water quality measures in addition to the data mentioned above. These include: • Water temperature • Suspended solids • Ammoniacal N • Nitrate N

Model Outputs	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
		- Nitrite N - Phosphate-P
Licensing Requirements	<i>The Wessex Water E. coli models for Warleigh Weir are not open models – they are restricted for use within the operations of Wessex Water. The data feeding into this model are flow (gained under open licence from the EA) and water quality data measured by Wessex Water.</i>	For commercial users, the RDMAI Open <i>E. coli</i> models require an Open Meteo API subscription to extract historical weather data.

2.2 Model Outputs

Table 2.2 Comparison of model outputs

Model Outputs	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
Data Format	<i>The results of the Wessex Water Warleigh Weir classifier model are public facing and are published on a website, informing bathing water safety at Warleigh Weir. The website works according to a traffic light system – no concentrations of <i>E. coli</i> are given, rather, a general indication of bathing water safety is provided, along with some measured data: temperature, flow and rainfall.</i>	The RDMAI Open <i>E. coli</i> estimator models produce concentrations of <i>E. coli</i> (CFU/100 ml), while the classifier produces a binary output of lower risk or higher risk for human exposure based on thresholds of <i>E. coli</i> concentration. These results can be extracted as a csv file. The notebooks provided in the GitHub repository also produce various graphs showing performance metrics.

Model Outputs	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
	<i>Focus of model is the end-user (public) experience, with simple actionable outputs.</i>	For the estimator, these include: • X-Y scatter plots • Histograms of error: absolute error; mean error by <i>E. coli</i> concentration bins; absolute mean estimation error by <i>E. coli</i> concentration bins. • Frequency histograms of simulated and observed <i>E. coli</i> distributions. For the classifier, these include • Confusion matrices, illustrating rates of false positives, false negatives, true positives and true negatives. RDMAI models have multiple options for output formats depending on end-user requirements.
Temporal Resolution	<i>The Wessex Water Warleigh Weir models give half-hour simulations.</i>	The temporal resolutions of the RDMAI Open <i>E. coli</i> models differ, and are ultimately dependent on the temporal resolutions of the input data: • Light model: dependent on the temporal resolutions of the Open Meteo weather data and MODIS remote sensing data – generally fine temporal resolution of a day or finer.

Model Outputs	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
		Advanced model: dependent on the temporal resolution of the supporting water quality data – if using EA water quality monitoring data, temporal resolution is typically 1 or 2 readings per month, depending on the monitoring point.
Geographic Constraints	<i>The Wessex Water model is specifically focused on Warleigh Weir – a popular summer bathing spot along the River Avon, near Bath.</i>	The training geographical extent of the RDMAI Open <i>E. coli</i> models depends on the variant: - The advanced model was trained on data limited to England due to the dependence on EA water quality data. - The light model was trained on data for Great Britain as the training data included <i>E. coli</i> measures by the Scottish and Welsh environment regulators. However, the training geographical extent does not limit where the model can be applied. A greater limitation of spatial context relates to input data: - The light model relies on land use, weather and MODIS remote sensing data, and can therefore be applied to any point.

Model Outputs	Wessex Water Warleigh Weir (Existing Industry Model)	RDMAI Model
		The advanced model is dependent on supporting water quality information and so is limited to points at which these water quality variables are measured.

2.3 Time and Cost Comparison

Preparing the input dataset and running the RDMAI *E. coli* light models for around 2000 data points takes about 4 hours, provided the model user is proficient. Preparing the dataset for the RDMAI *E. coli* advanced models can take longer, particularly if the supporting water quality data need to be extracted from the EA water quality monitoring database (WIMS) and processed. For a 2000-point dataset, this can increase the time required to process the data and run the model to at least 6 to 8 hours. Much of this time is due to the time taken to extract weather and remote sensing data – this is automated and can be run in the background. The only direct cost of applying the RDMAI *E. coli* models is applicable if the model is used by a commercial organisation - a professional subscription is needed to access Open Meteo API historical weather data, which costs €99/month. It should be noted that the benchmarking was performed on a local machine (laptop) without accessing cloud computing. Some organisations may make use of cloud computing, which will have additional associated costs.

The cost of water sample testing for *E. coli* detection and/or quantification will vary depending on method used and between laboratories. Environmental water quality tests are typically conducted in-house by respective water companies. Independent laboratories may charge between £20 - £100 per sample for coliform/*E.coli* testing (culture-based standard tests such as membrane lactose glucuronide agar - MLGA - methods), with costs linked to scale of order. There may also be an associated increase in costs if a dilution series is required for quantification, which would be applicable to faecal contaminated waters with very high concentrations of *E. coli*. The EA conducted around 12,000 *E. coli* sample measurements between July 2024 and July 2025 at 1029 locations in England. Given the costs per sample processed given above, the annual costs to the EA to conduct bathing water quality assessments for *E. coli* (considering just laboratory sample processing, not labour or transport costs) could range between £240,000 and £1.2 Million. Given that the EA classifies bathing water quality based on samples for the previous four years, *E. coli* sample processing costs associated with bathing water classification could range between £960,000 and £4.8 Million.

The time taken to receive sample results will vary depending on method of analysis used. Presumptive laboratory results for each *E. coli* measurement on MLGA agar are available after incubation for at least 18 hours, if further confirmation testing is required then additional reagents and incubation for 21 ± 3 hours will be necessary. IDEXX Colilert®-18 tests offer a faster result turnaround with confirmed *E. coli* results being available within 18-22 hours after analysis. These times are not withstanding the duration of travel between sample collection and reception at the laboratory for analysis which must be < 24 hours to meet Environment Agency standards.

Table 2.3 shows a time and cost comparison between the RDMAI Open *E. coli* model and traditional laboratory testing to generate a dataset of 2,000 *E. coli* data points. The cost of running the RDMAI Open *E. coli* model is free for non-commercial organisations or as little as 0.04% of traditional laboratory measurements, saving between £40,000 to £200,000, whereas the time saved by applying the RDMAI Open *E. coli* model can reach 48,000 hours (2,000 days).

Table 2.3 Time and cost comparison between the RDMAI Open *E. coli* model and typical laboratory measurements for generating a sample dataset of 2000 *E. coli* data points

Criteria	RDMAI Open <i>E. coli</i> model	Laboratory measurements
Time (h)	4 to 8 hours, depending on whether the light or advanced model is applied.	Between 18 to 24 h per test, totalling 36,000 to 48,000 hours.
Cost (£)	0£ for a non-commercial organisation, £86 (€99) per year for a commercial organisation.	Between £40,000 and £200,000

2.4 Model Performance

2.4.1 In comparison to Wessex Water Warleigh Weir model

Figure A1, A2, A3 and A4 in Appendix A show the distribution of land use, weather and MODIS input data for both the light and advanced models applied to the Wessex Water Warleigh Weir data. The single site of the dataset is reflected in the land cover and distance to coast, which are constant for all data lines. The remaining variables show variation as they change over time. Observed *E. coli* mostly fall between 0 and 5,000 CFU/100 ml, but observations reach as high as 30,000 CFU/100 ml. Most *E. coli* observations at Warleigh Weir are, as expected, during the daylight hours. The graphs illustrate the range of weather-related variables used as input to the model, including precipitation, temperature, wind speed, sunshine duration, and evaporation. These weather-related variables include sub-daily (such as precipitation at 11 am), daily (such as the daily sum of precipitation) and antecedent (such as the sum of rainfall over the preceding 7 days) measures.

2.4.2 Wessex Water Warleigh Weir Model

Wessex Water has both a regressor (estimator) and classifier model for Warleigh Weir. The public facing data is generated by the classifier.

Performance of the regressor (estimator)

As shown in Figure 2.1**Error! Reference source not found.**, the Wessex Water model seems to over-predict lower concentrations and slightly under-predict higher concentrations, but most of the scatter is near the 1:1 line. The results shown in Figure 2.1**Error! Reference source not found.** are further emphasised in Figure 2.2**Error! Reference source not found.**. This shows a confusion matrix, with an *E. coli* threshold of 500 CFU/100 ml applied. The figure shows that the Wessex Water model has a false positive (FP) rate of 95% and a false negative (FN) rate of 0%. This means that the model successfully predicts all occasions when *E. coli* exceeds a threshold of 500 CFU/100 ml. However, it also tends to be conservative, with a high FP rate.

Results for the classifier

Wessex Water uses thresholds in their classifier for Warleigh Weir that are different to those implemented in the RDMAI Open *E. coli* classifier model. The threshold was initially 1,000 CFU/100 ml but was then reduced to *E. coli* 900 CFU/100 ml. The dataset provided by Wessex Water provides both the measured *E. coli* and the classifier results, shown as a binary variable. These data were analysed to assess the FPs and FNs for the Warleigh Weir classifier according to their own thresholds.

The results showed:

- FP: 47.6%
- FN: 0.31%

Figure 2.1 X-Y scatterplot of predicted versus observed *E. coli* concentration (CFU/100 ml) by the Wessex Water Warleigh Weir model

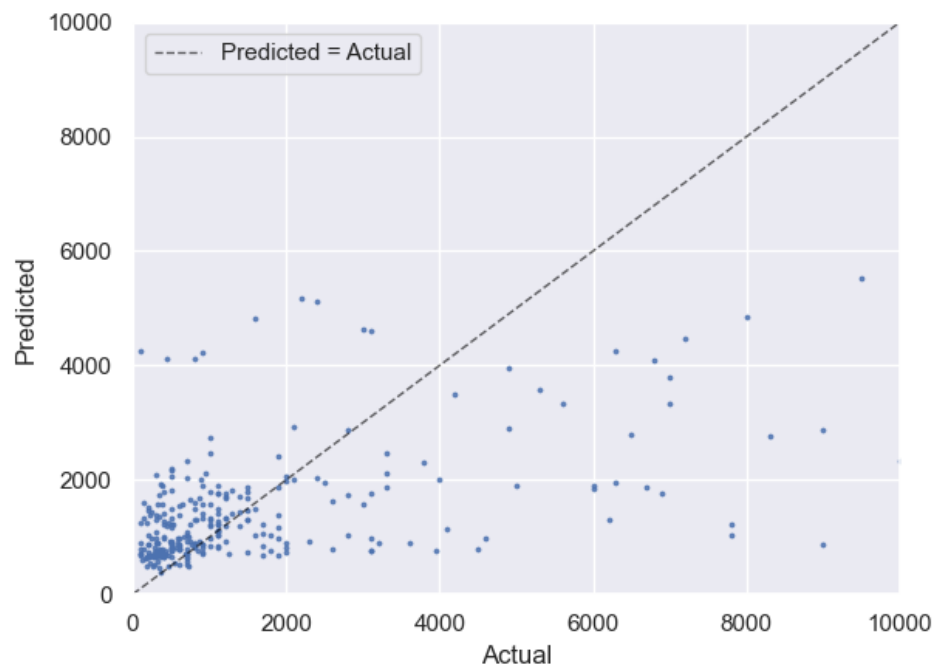
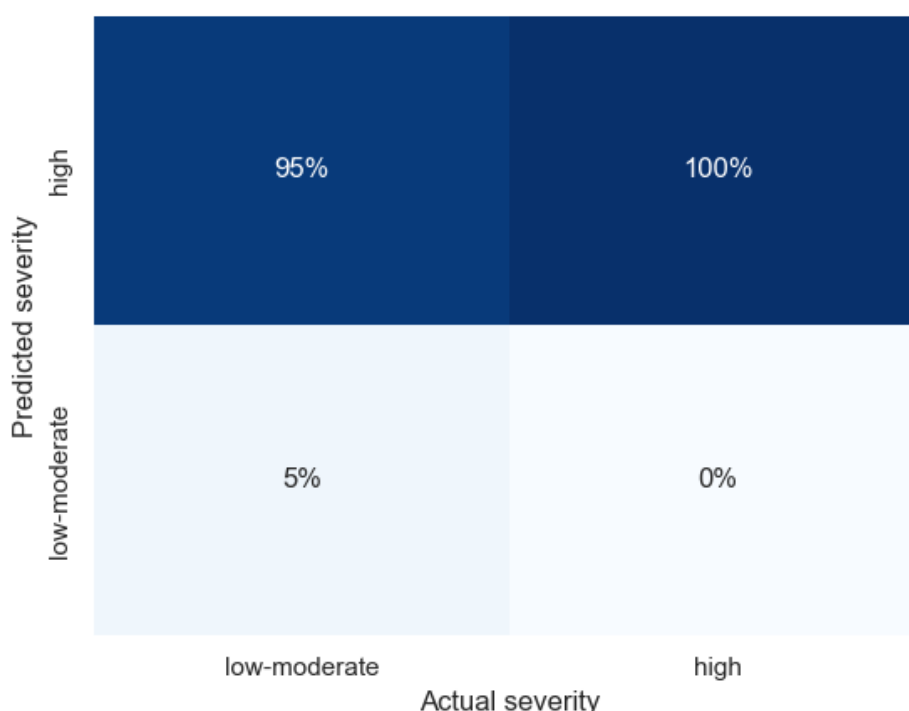


Figure 2.2 Confusion matrix for the Wessex Water Warleigh Weir *E. coli* model with a threshold of 500 CFU/100 ml applied



2.4.3 RDMAI Open *E. coli* light model

Estimator model

As shown in Figure 2.3**Error! Reference source not found.**, the performance of the light estimator model in predicting actual *E. coli* concentrations was limited. The X-Y scatter plot shows that the estimator tended to underestimate high concentrations.

This is illustrated in the confusion matrix in Figure 2.4**Error! Reference source not found.**, **Error! Reference source not found.** where a threshold of 500 CFU/100 ml is applied to the estimated *E. coli* concentrations to evaluate the model's ability to classify risk. The figure shows that the model has a FP rate of 25% and a FN rate of 51% at that threshold.

Classifier model

As shown in Figure 2.5, when applying the RDMAI Open *E. coli* light classifier to the Wessex Water dataset, the model had a 61% FN ($185/301 \times 100$) and a < 1% FP ($2/301 \times 100$).

Figure 2.3 RDMAI Open E. coli light estimator predicted E. coli concentrations versus observed E. coli concentrations

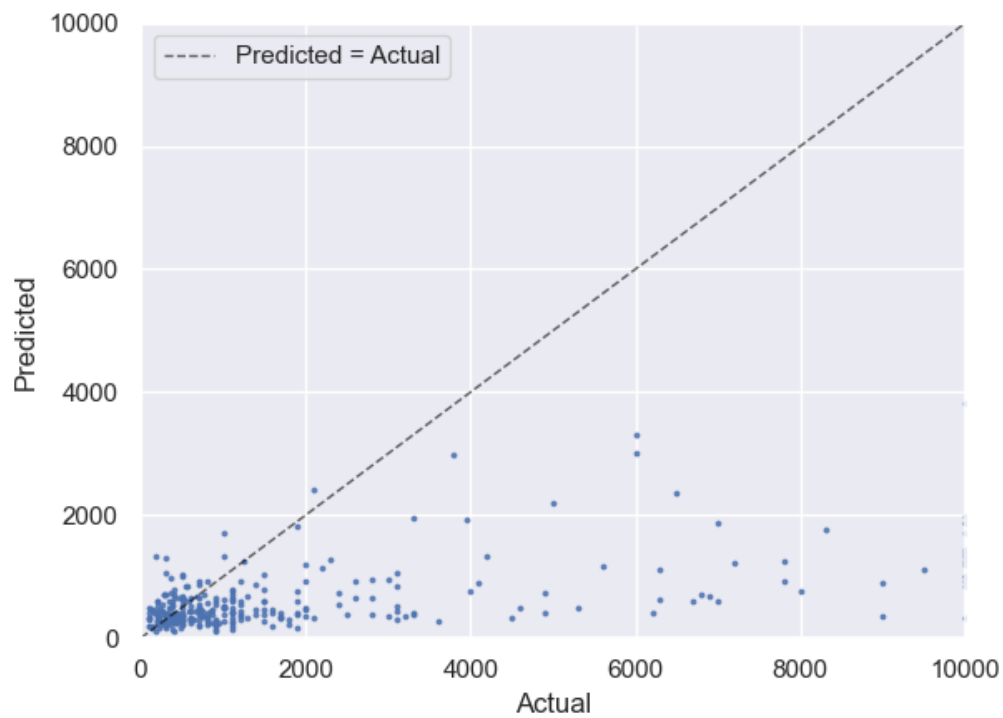


Figure 2.4 Confusion matrix for the RDMAI Open E. coli light estimator with a threshold of 500 CFU/100 ml applied

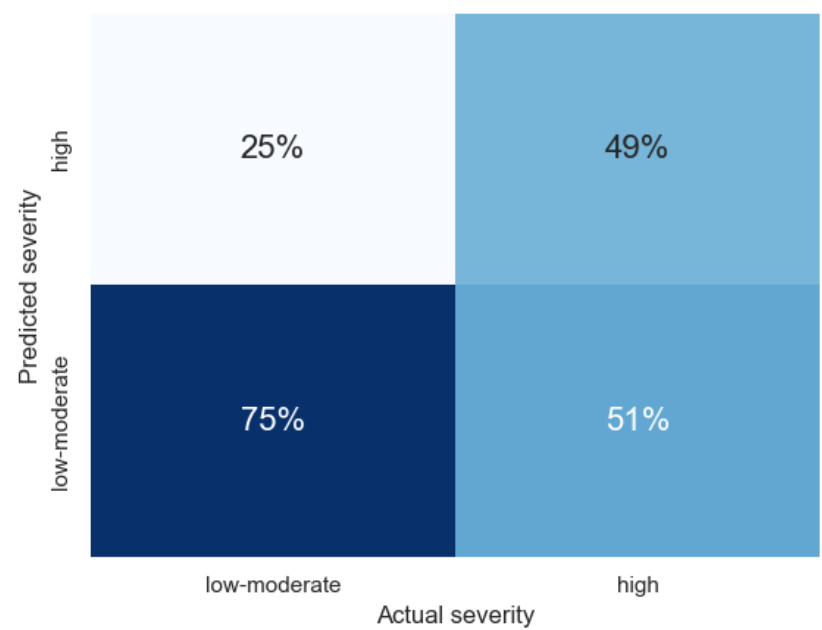
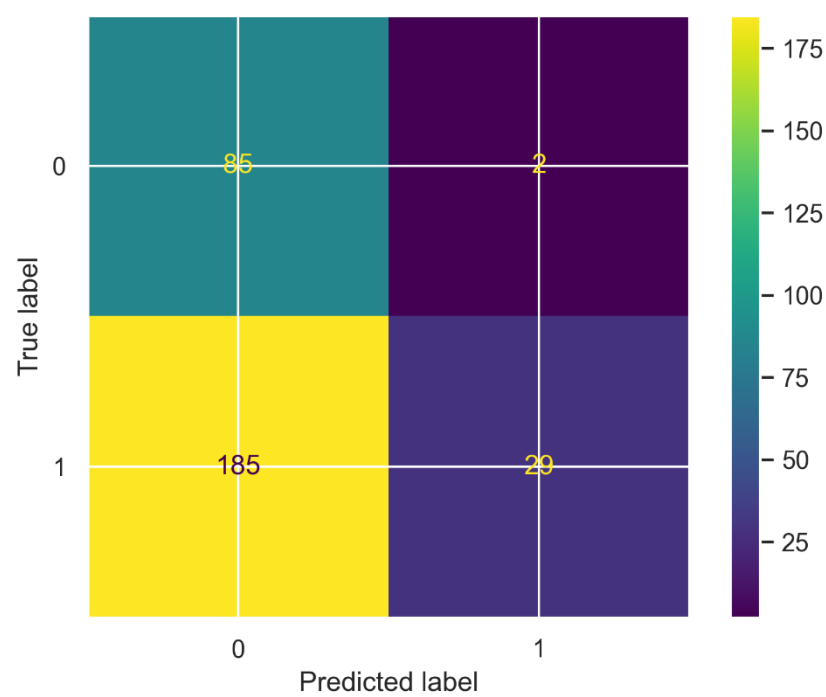


Figure 2.5 Confusion matrix of the RDMAI Open E. coli light classifier model with a threshold of 500 CFU/100 ml applied



RDMAI Open *E. coli* advanced model

Estimator model

Figure 2.6 **Error! Reference source not found.** shows that the performance of the advanced estimator model in predicting observed *E. coli* concentrations was limited, with concentrations consistently underestimated. For this reason, the predicted concentrations were split based on a threshold of 500 CFU/100 ml to further assess the ability of the model to classify risk. The resulting confusion matrix shown in Figure 2.7 also illustrates the observations from the predictions of the estimator. This figure shows that the model has a FN rate of 66% and a FP rate of 10%.

Classifier model

As shown in the confusion matrix in Figure 2.8, the advanced classifier model, when a threshold of 500 CFU/100 ml is applied, generated 5% FP ($9/176 \times 100$) and 37% FN ($65/176 \times 100$).

Figure 2.6 RDMAI Open *E. coli* advanced estimator predicted *E. coli* concentrations versus observed *E. coli* concentrations

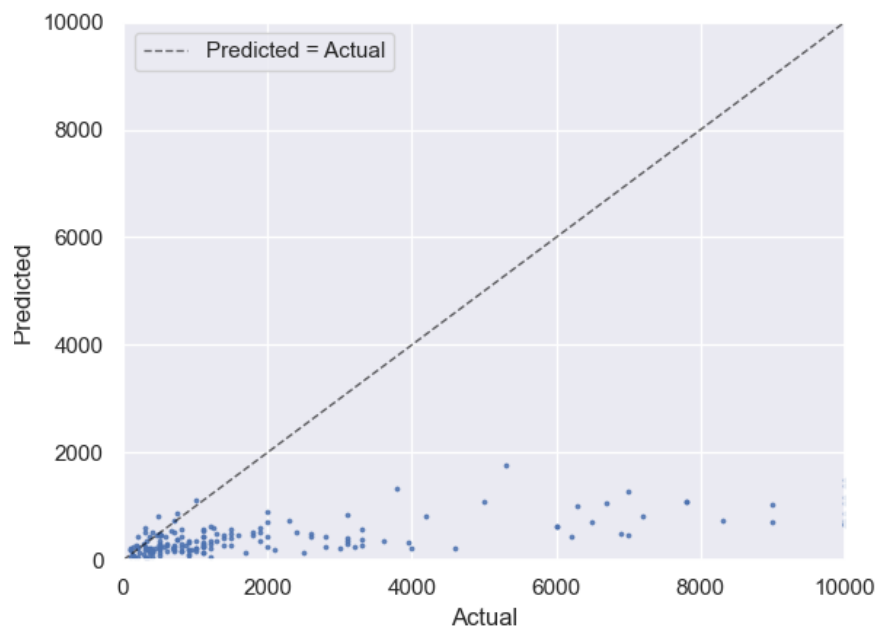


Figure 2.7 Confusion matrix for the RDMAI Open *E. coli* advanced estimator with a threshold of 500 CFU/100 ml applied

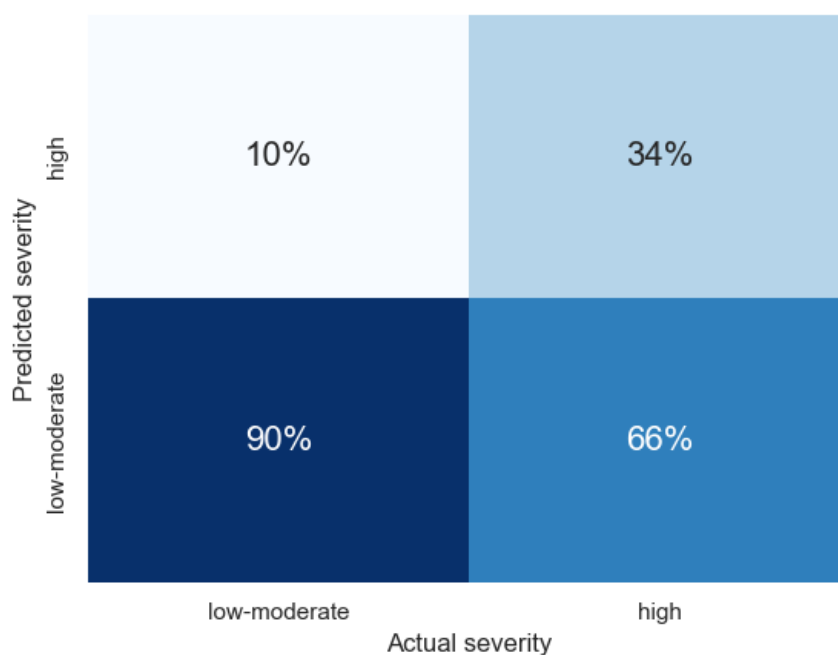
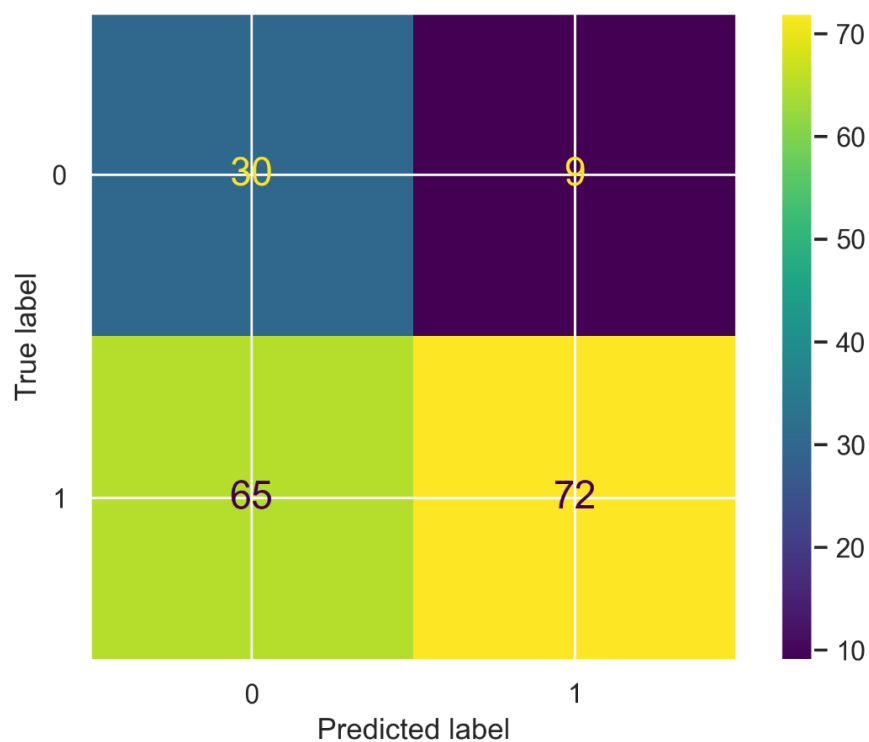


Figure 2.8 Confusion matrix of the RDMAI Open *E. coli* advanced classifier model with a threshold of 500 CFU/100 ml applied.



2.4.4 In comparison to citizen science *E. coli* measured data

Figure A5, A6, A7 and A8 in Appendix A show the distribution of input data applied to the RDMAI Open *E. coli* light models. The points are based on citizen science *E. coli* monitoring, with supporting land use, weather and remote sensing data added to the dataset for those points. In contrast to the Wessex Water Warleigh Weir dataset, this dataset has many more points, including coastal and inland points, and associated differences in land cover. The graphs show that agricultural land dominate the buffered area around the monitored points, followed by grassland and built-up land. Time is constant as 12:00 as time of observations were not indicated, and a measurement time of midday was assumed. Observed concentrations of *E. coli* are mostly concentrated between 0 and around 2,000 CFU/100 ml, but reach as high as 10,000 CFU/100 ml.

2.4.5 RDMAI Open *E. coli* light model

Estimator model

As shown in Figure 2.9, the performance of the RDMAI Open *E. coli* light estimator in representing the citizen science dataset of *E. coli* measures was limited. However, in contrast to the Warleigh Weir dataset, in which higher *E. coli* concentrations were consistently under-simulated, the results for this dataset show a wider scatter above and below the 1:1 line. The 'column' of scatter points seen at around 1000 CFU/100 ml is due to many observations being stated as 'greater than' 1000 CFU/100 ml, which were taken to be 1000 CFU/100 ml.

Figure 2.10 shows the confusion matrix resulting from the results of the RDMAI Open *E. coli* light estimator when a threshold of 500 CFU/100 ml is applied to the predicted concentrations to evaluate the model's ability to classify risk. The results show a FP rate of 57% and a FN rate of 26%.

Classifier model

Figure 2.11 shows the confusion matrix generated from the results of the RDMAI Open *E. coli* light classifier when a threshold of 500 CFU/100 ml is applied. FNs are 19% ($400/2100 \times 100$), whereas FPs are 14% ($287/2100 \times 100$).

Figure 2.9 RDMAI Open *E. coli* light estimator predicted *E. coli* concentrations versus observed *E. coli* concentrations for the citizen science dataset

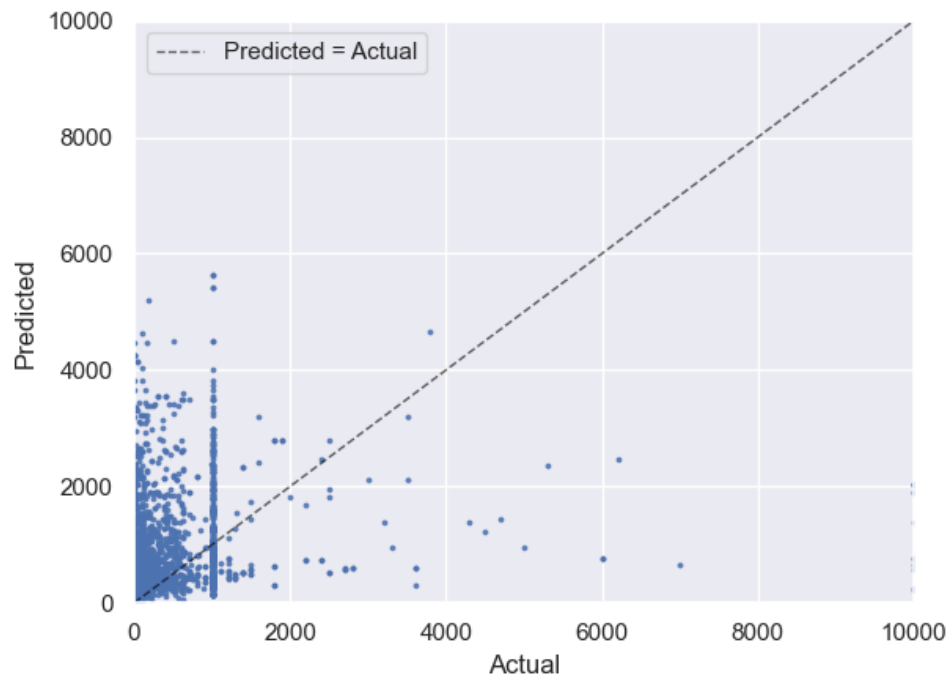


Figure 2.10 Confusion matrix for the RDMAI Open *E. coli* light estimator applied to the citizen science dataset with a threshold of 500 CFU/100 ml applied

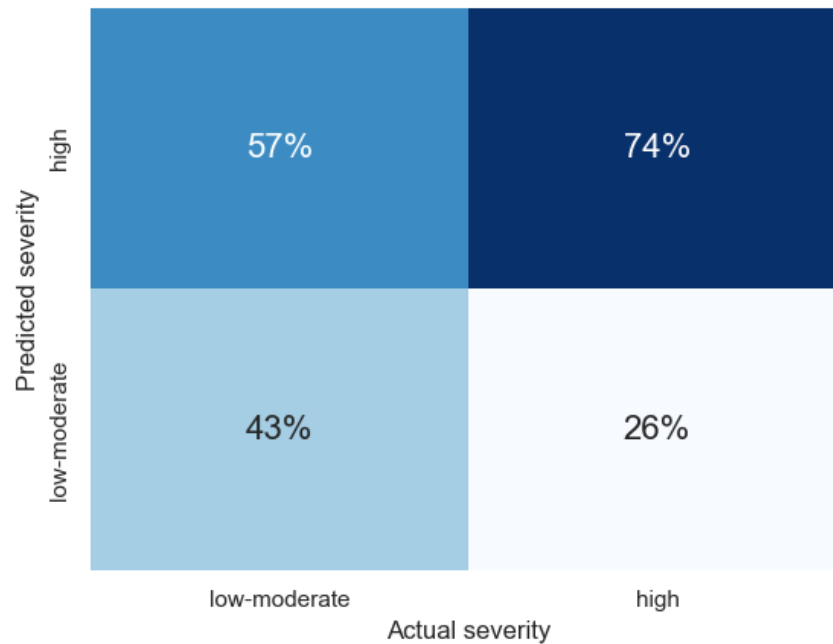
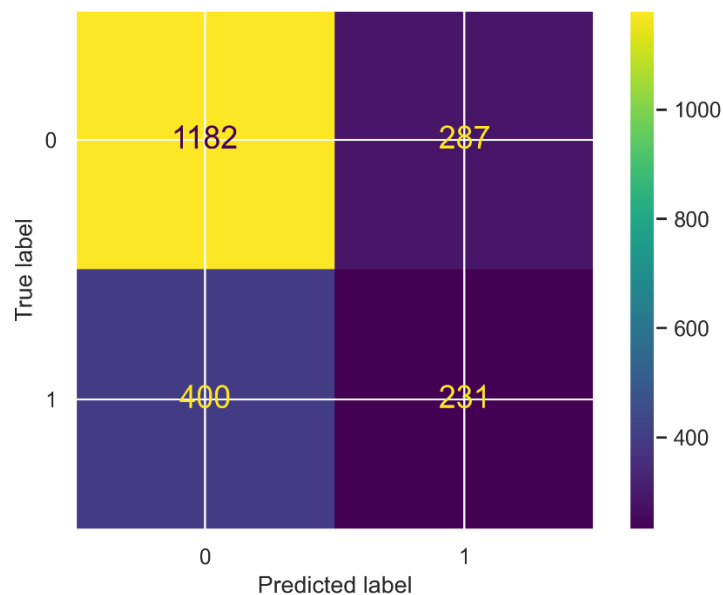


Figure 2.11 Confusion matrix of the RDMAI Open *E. coli* light classifier model applied to the citizen science dataset with a threshold of 500 CFU/100 ml.



2.5 Model Performance Summary Statistics

The results shown in Figure 2.2Error! Reference source not found. indicate that the Wessex Water Warleigh Weir model is accurate (with the given actual data) in identifying when concentrations exceed thresholds of concern, with 0% FNs. It is conservative, in that the regressor (estimator) has a high proportion of FPs (95%). However, it should be noted that these results are according to the imposed threshold of 500 CFUs/100 ml. The classifier, trained according to Wessex Water's thresholds of 900 to 1000 CFUs/100 ml, obtains a FP rate of 45% and no FNs.

The RDMAI Open *E. coli* light estimator, applied to the Warleigh Weir data, tends to underestimate high *E. coli* concentrations. Consequently, under a threshold of 500 CFU/100 ml, the model has a high FN rate of 51%. The RDMAI Open *E. coli* light classifier also had a high rate of FNs of 61%. The RDMAI Open *E. coli* advanced estimator and classifier applied to Warleigh Weir similarly tended to underestimate *E. coli* and give high rates of FNs. However, the 37% FN rate under the RDMAI Open *E. coli* advanced classifier was the lowest FN of all models.

For the citizen science *E. coli* data, FNs under the RDMAI Open *E. coli* light estimator and classifier were still high, at 26% and 19%, respectively. However, they were improved compared to the result for the model applied to the Warleigh Weir data. The limitation of the citizen science dataset of many observations being stated as 'greater than' 1000 CFU/100 ml might have contributed to the FPs of the RDMAI Open *E. coli* light models (Figure 2.10, Figure 2.11).

Table 2.4 Summary performance metrics of the RDMAI Open *E. coli* model benchmarking

Model	R ²	RMSE (CFU/100 ml)	MAE (CFU/100ml)	FP (%)	FN (%)
Warleigh Weir dataset					
Wessex Water Warleigh Weir regressor model (FPs and FNs estimated according to a 500 CFU/100 ml threshold)	0.2	3907	1702	95%	0%
Wessex Water Warleigh Weir classifier model (FPs and FNs estimated according to their own thresholds – 900 to 1,000 CFU/100 ml threshold)	N/A	N/A	N/A	48%	0%
RDMAI Open <i>E. coli</i> light estimator	-0.085	2752	1465	25%	51%
RDMAI Open <i>E. coli</i> light classifier	N/A	N/A	N/A	< 1%	61%
RDMAI Open <i>E. coli</i> advanced estimator	-0.286	3110	1812	10%	66%
RDMAI Open <i>E. coli</i> advanced classifier	N/A	N/A	N/A	5%	37%
Citizen science dataset					
RDMAI Open <i>E. coli</i> light estimator	-0.73	1250	761	57%	26%
RDMAI Open <i>E. coli</i> light classifier	N/A	N/A	N/A	14%	19%

3. Benchmarking Conclusion

The benchmarking exercise against the Wessex Water Warleigh Weir model faced a significant limitation: the Wessex Water model is a site-specific model, trained on data for Warleigh Weir. In contrast, the RDMAI Open *E. coli* models are national-level models. Therefore, it is expected that the Wessex Water model would achieve better results for Warleigh Weir. The model performance statistics should be considered within this context.

The benchmarking of the RDMAI Open *E. coli* models shows that the models at this stage have relatively limited accuracy – performance was limited when benchmarked with a site-specific model (Wessex Water Warleigh Weir model) but improved somewhat when compared to a citizen science dataset for around 84 sites across Great Britain). Focusing on the false negatives, which is arguably the most important metric for a risk assessment model, the RDMAI Open *E. coli* models produced false negatives of between 37% and 66% for the Warleigh Weir dataset. They showed an improved performance on the citizen science dataset with false negatives of 19% to 26%. It should be noted that the Wessex Weir dataset contained a relatively high proportion (6%) of high *E. coli* measures exceeding 10,000 CFU/100 ml. Since the training *E. coli* concentration data for the RDMAI Open *E. coli* model was capped at 10,000 CFU/100 ml, this might have had an impact on the model performance against the Wessex Weir dataset. The citizen science dataset in contrast had a lower proportion of extreme high *E. coli* measures exceeding 10,000 CFU/100 ml (<1%).

The bias of the models to the majority class (low *E. coli* measures) results in under simulation of high *E. coli* concentrations, and a high rate of false negatives. The RDMAI Open *E. coli* models perform better when applied nationally (to the citizen science data) than to an individual point (Warleigh Weir). While the RDMAI Open *E. coli* models are not sufficiently accurate to replace current water company and EA site-specific models, there is potential for their application alongside citizen science data and site-specific models.

There is potential to improve the RDMAI Open *E. coli* model, given that it is open source and accuracy will be dependent on the availability of data to train. This, in combination with its wide spatiotemporal application potential and low cost, make it an attractive option to pursue and improve. The comparison of cost and time between the RDMAI Open *E. coli* model and traditional laboratory *E. coli* measurements shows potential money and time savings of as much as £200,000 and 6,000 days, respectively, for a 2,000-point dataset. The RDMAI Open *E. coli* model has the potential to act as a foundation for a site-specific model, such as the Wessex Water model. It would need some improvements in feature engineering, such as accounting for the upstream contributing catchment, and possibly a bias towards being more conservative, i.e., reducing false negatives as much as possible.

Appendix A

Figure A.1 Distribution of input data for the RDMAI Open *E. coli* light model

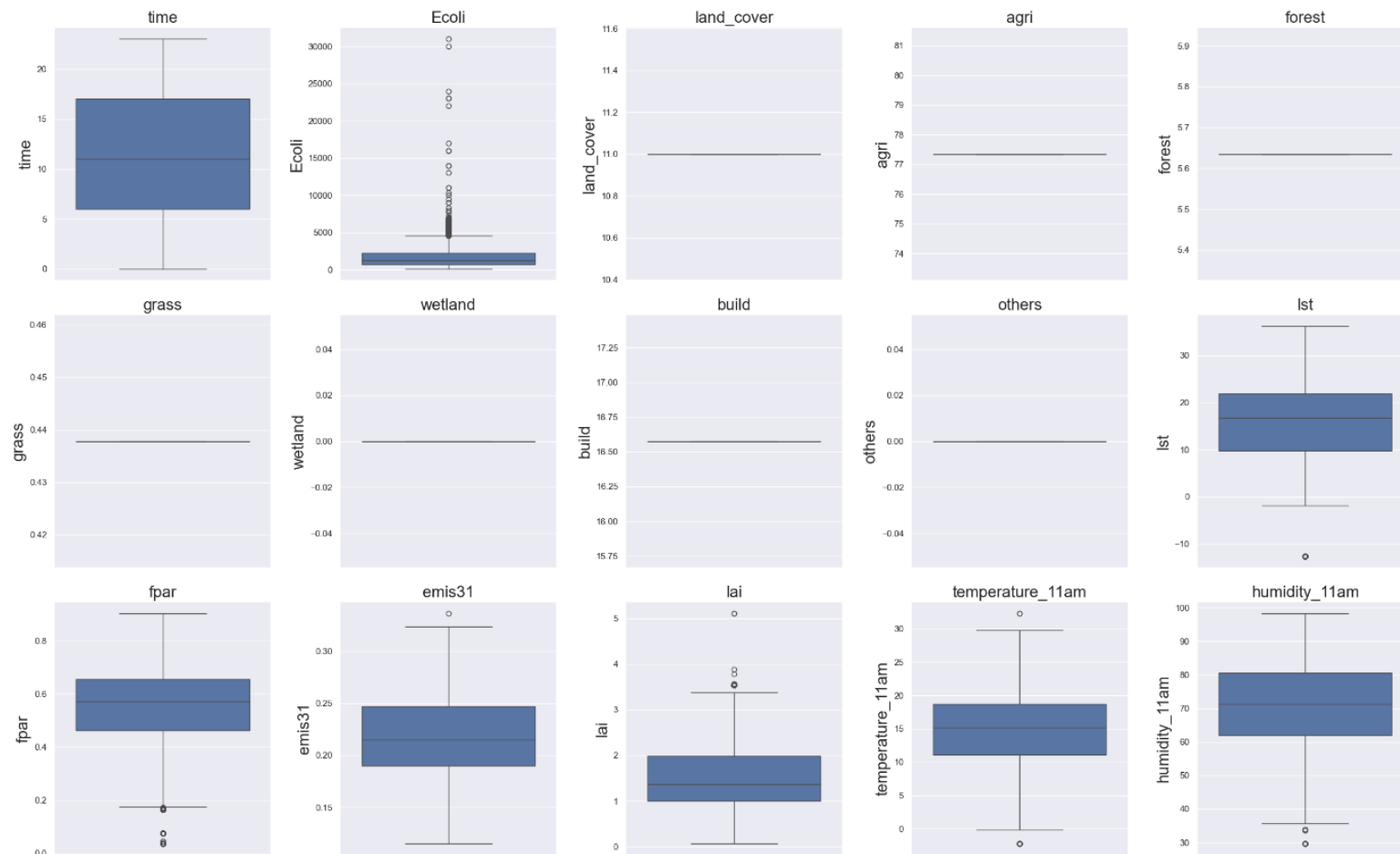


Figure A2 Distribution of input data for the RDMAI Open *E. coli* light model

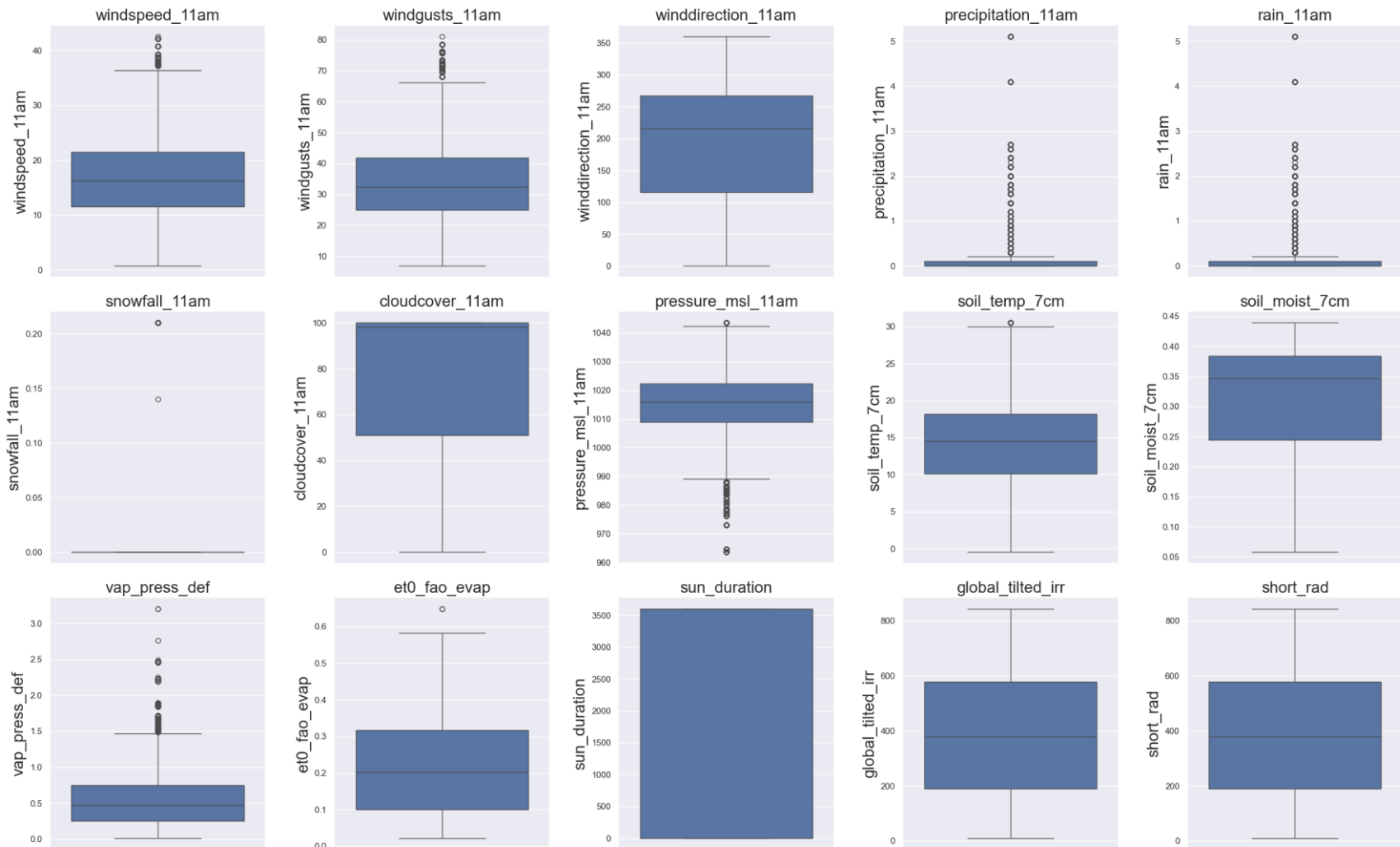


Figure A3 Distribution of input data for the RDMAI Open *E. coli* light model

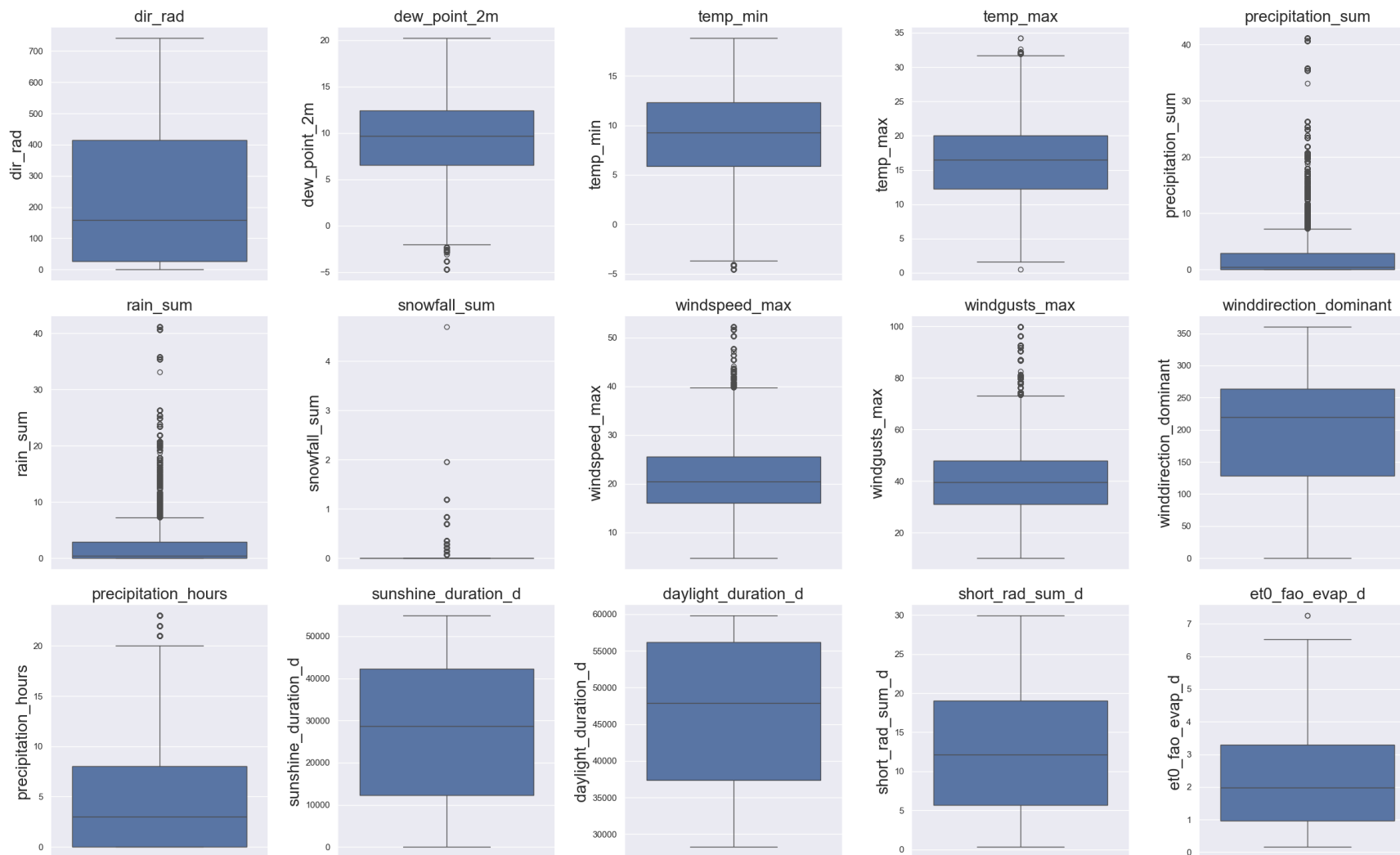


Figure A4 Distribution of input data for the RDMAI Open *E. coli* light model

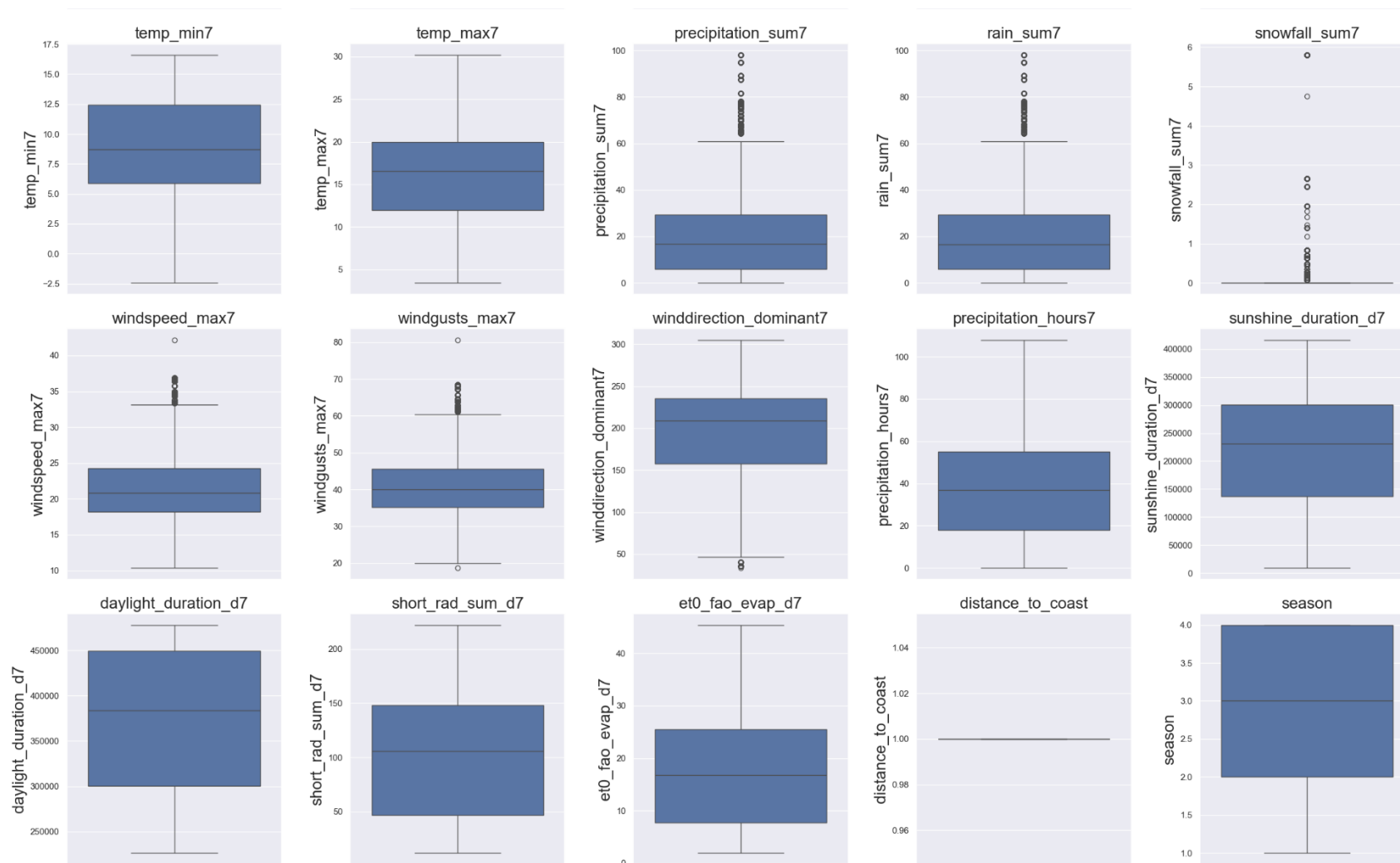


Figure A5 Distribution of data applied to the RDMAI Open *E. coli* light models based on citizen science *E. coli* monitoring points

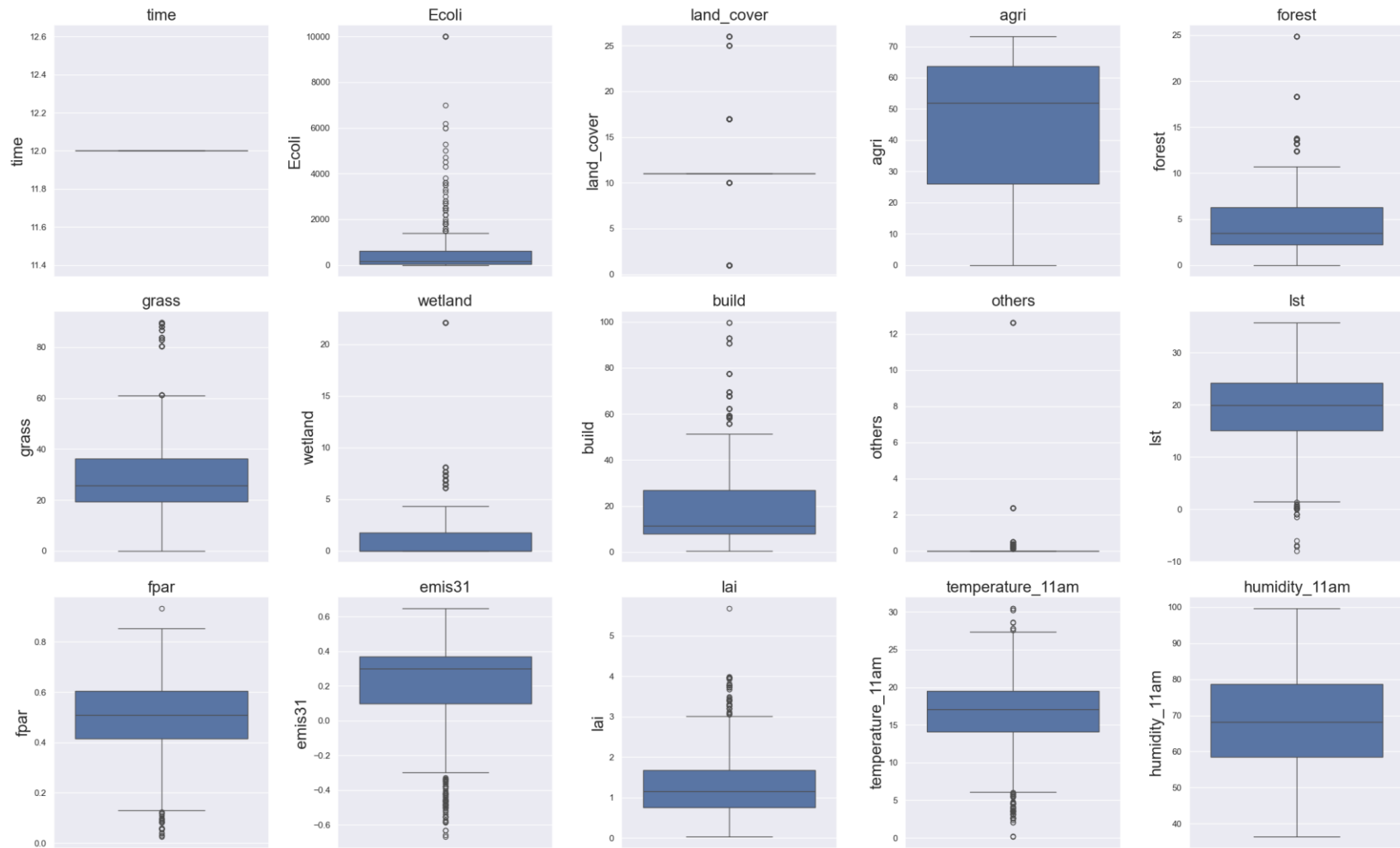


Figure A6 Distribution of data applied to the RDMAI Open *E. coli* light models based on citizen science *E. coli* monitoring points

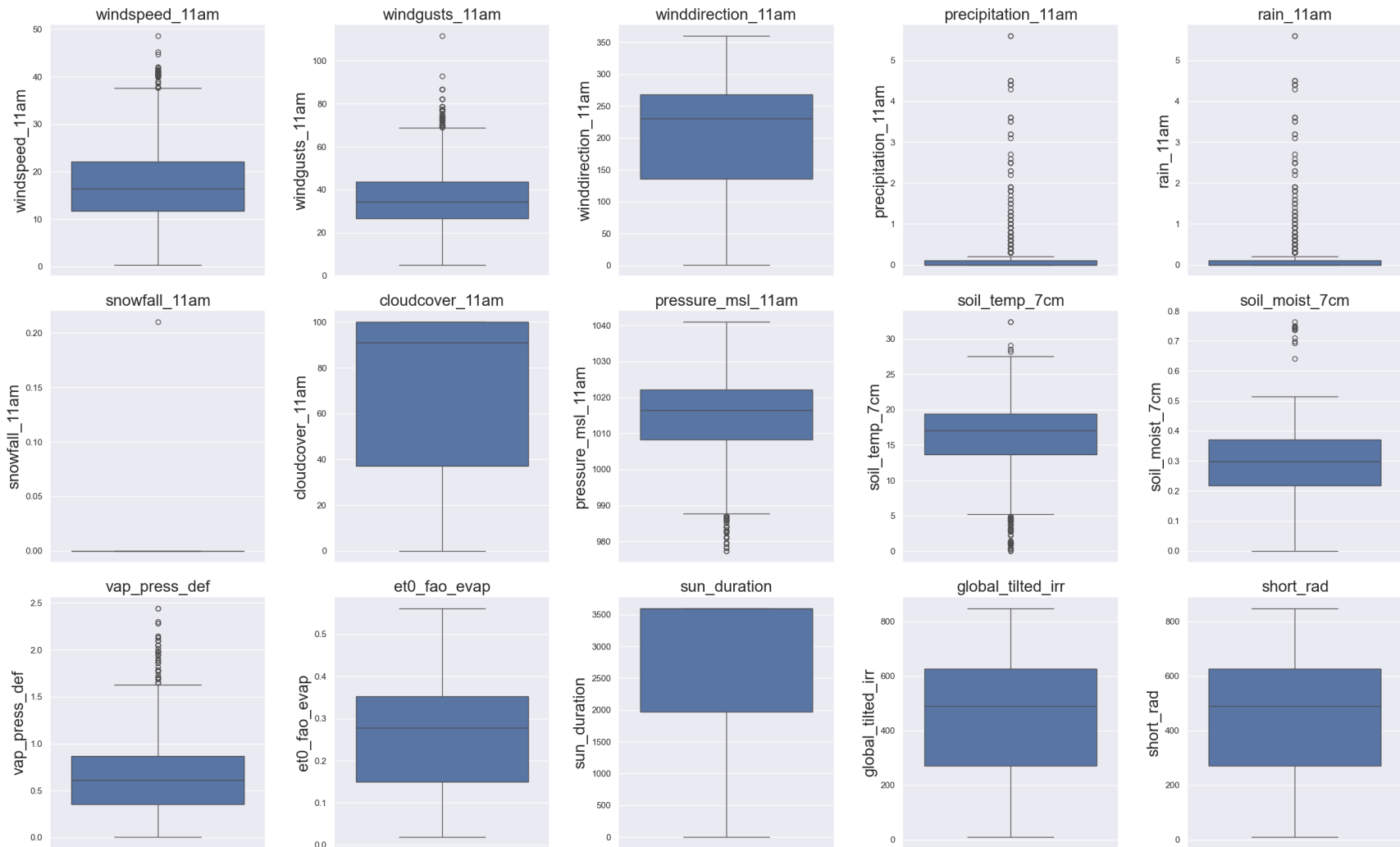


Figure A7 Distribution of data applied to the RDMAI Open *E. coli* light models based on citizen science *E. coli* monitoring points

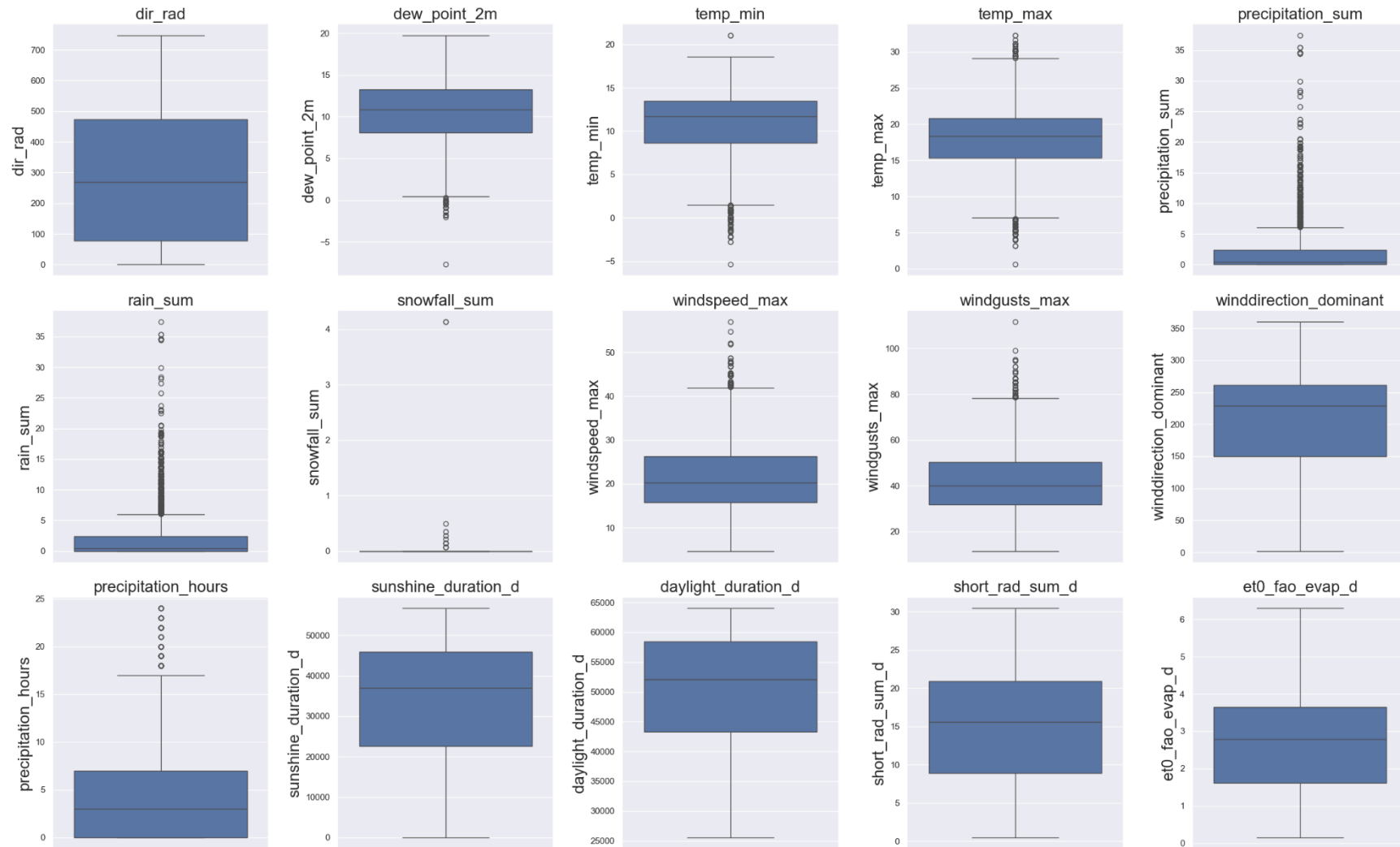
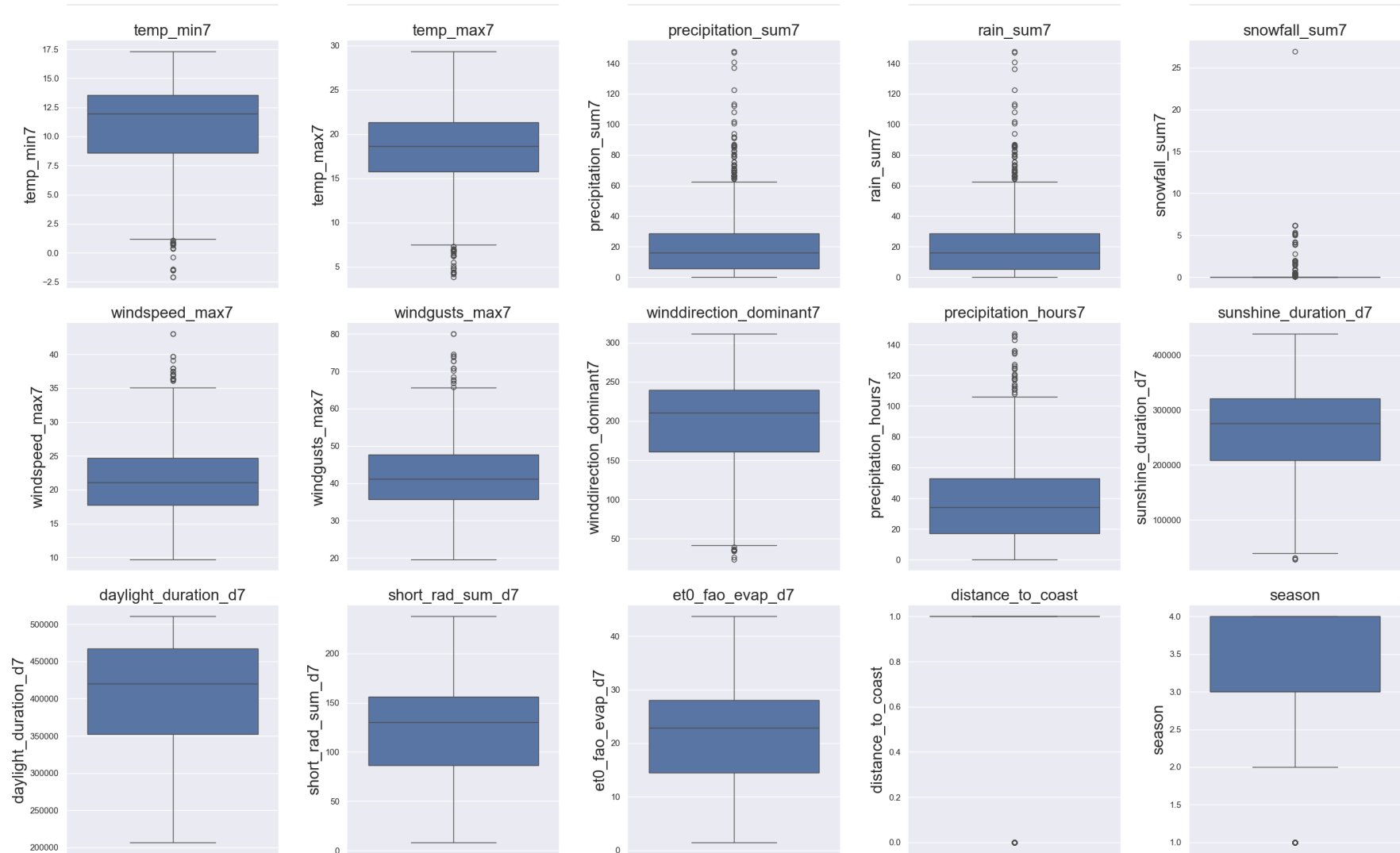


Figure A8 Distribution of data applied to the RDMAI Open *E. coli* light models based on citizen science *E. coli* monitoring points



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